THE COTTON LINE — Development Log

Hey! So this right here is my Development Log. To be completely honest, I maintain a log like this for pretty much every research project I work on. Initially, I just started doing this for my own personal benefit—mostly so I wouldn't lose my mind trying to remember where I left things off the night before, or what exact dataset was sitting on my disk.

But for this project, I chose to write it down as a proper, ongoing story. I wanted to narrate the actual execution of the project step-by-step, just in case anyone else wants to see how the whole thing came together. I have mentioned all of my core reasonings, my sudden dataset changes, and even those annoying coding bugs and debugging sessions where things completely broke down before I fixed them. It is basically the raw, behind-the-scenes diary of how this project actually got built!

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Entry 1

I started pulling data for this region back in late May. My working name back then was Project Prism—had this clunky acronym (Predictive Regional Intelligence and Sustainable Management) that I eventually threw out. Honestly, I was just using a broad "climate-resilient Vidarbha" framing and had no fixed research question yet. I just had a strong gut feeling that I needed to build something on this specific area, so I did a massive first pass to grab every dataset a regional study would ever need. Before I actually pick a real hypothesis, I want to

list down exactly what is sitting on my drive right now (a lot of this was collected before I even knew what my paper was about, and I really don't want to lose track of where it all came from).

Here is what I already collected before the pivot

• Administrative boundaries — Got taluka-level shapefiles for all 11 Vidarbha districts (Akola, Amravati, Bhandara, Buldhana, Chandrapur, Gadchiroli, Gondia, Nagpur, Wardha, Washim, Yavatmal) from IIT Bombay's POCRA Maharashtra Census shapefile portal. Also saved the 1994 Development Boards Order — needed it for primary-source context on the region's backwardness and structural gaps. • Climate — Downloaded Landsat 8/9 Collection 2 Level-2 thermal band (Band 10, surface temp) scenes. Covers the whole footprint. Did a rough merge for the land-surface-temperature raster map too. • Agriculture — Got the district-wise Area-Production-Yield (APY) sheets for 2024-25, plus Cotton Corporation of India MSP procurement stats (2019-20 to 2024-25). Saved the Maharashtra Economic Survey 2025-26 agri chapters while I was at it. • Hydrology — Clipped the WorldClim 2.1 monthly rain data (10 arc-minute global grid) to the Vidarbha border polygon. • Infrastructure — Road/rail network vector lines from OpenStreetMap. • Terrain — SRTM 30-meter elevation tiles for the full grid. Merged, clipped, reprojected, and ran a quick slope calculation from the DEM. • Vegetation — Landsat 8/9 Level-2 red/near-infrared bands (Band 4 and Band 5) matching the climate scenes. Need this for the NDVI formula later.

Two categories are still completely empty on my computer: **Land_use** and **Economics**. Forest Rights Act data —things like claims filed, titles given out, and community forest areas—just does not exist anywhere as a clean, ready-to-download district list. What is actually out there is a total mess scattered across old parliamentary answers, CAG audit reports, and single district-office PDF notices instead of one useful file. That is still an open problem that I have not solved yet.

Today's addition — Population. I downloaded village-level Census shapefiles for all 11 districts. I used the same IIT Bombay POCRA portal that I already went to for my admin borders. The great thing is this version carries over 600 census facts per village (like population counts, literacy, sex ratio, and other numbers), not just the basic map lines. All 11 district files are sitting on my disk now.

The pivot. My original "climate-resilient Vidarbha" idea was way too broad. It felt too close to what everyone making a regional satellite project already does. You know the usual stuff—land monitoring, vulnerability indexes, and the exact same tools I already used across the rest of my work portfolio. What I actually want for this project is something unique that could only come out of my own academic background (Economics, Political Science, History, and Geography) rather than just a neat, technically okay satellite analysis.

The new idea: Vidarbha's own old history gives me a great pair to test out. The five western districts (Amravati, Akola, Buldhana, Washim, and Yavatmal) used to be the old province of Berar. The British took this land directly in 1853 and set up the Ryotwari farming system. They forced farmers to grow only cotton because the American Civil War cut off the normal cotton supply going to Lancashire.

On the other side, the six eastern districts (Nagpur, Wardha, Chandrapur, Bhandara, Gondia, and Gadchiroli) were the core of the Central Provinces. They had a completely different setup called the Malguzari system. That old border was officially wiped out at Independence. But the crazy thing is that it runs almost exactly along the modern Amravati Division and Nagpur Division line (which is a boundary I already had map files for on my computer). It also matches the exact map of the 1900 Berar famine (which hit the heavy cotton areas worst) and fits today's farming trouble map of Vidarbha, which is still worst in those exact same western districts.

My new project name is: **THE COTTON LINE**. My core question is simple: does that 150-year-old colonial border still show up as a measurable spatial jump today in cotton dependency, drought trouble, and farmer stress? I am checking this directly against data I get from satellites rather than just guessing from old history books.

I will put the full theory details on its own dashboard page later. This includes institutional persistence theory, the historical reasons why Berar grew cotton, and the physics behind the satellite numbers I use. It will look like the other theory pages in my portfolio instead of sitting as a separate, boring standalone document right now.

Things I still need to finish after this entry:

- Land_use / LULC data — I have not found or sourced this yet.
- Economics category — I still do not have a separate source for this (I might just mix it into the Agriculture MSP and Economic Survey data that I already have on my computer).
- FRA data — I found no clean file anywhere online. I will probably have to type this out by hand using the scattered PDF pages instead of a single easy download.
- The historic Berar and Central Provinces border line — I still need to map and check this properly against my modern Amravati and Nagpur Division files. Books say they look "roughly the same" (but I have not proved that they are exactly identical on a map yet).
- I have not run any math or analysis yet. This whole entry is just me checking my files and changing my project idea.

Entry 2

I finally made the first real map lines for this project. At this point, I changed my main folder name from Project Prism to THE_COTTON_LINE. So from now on, everything uses this new folder name.

What I did. I opened the taluka map layer on my computer, which has 120 talukas across all 11 districts that I got earlier. Then I split them into two old historic groups by their district names. For the old Berar Ryotwari side, I picked the Amravati Division districts (Akola, Amravati, Buldhana, Washim, and Yavatmal). For the old Central Provinces Malguzari side, I picked the Nagpur Division districts (Nagpur, Wardha, Bhandara, Gondia, Chandrapur, and Gadchiroli). I used a dissolve tool to merge each group of talukas into one big shape per division (56 talukas became the Amravati Division shape, and 64 talukas became the Nagpur Division shape). After that, I grabbed the shared border edge where these two big shapes touch to get my modern line.

That border line looked like a complete mess at first. It came out as 256 tiny broken line pieces. This was not a coding error. It just happened because the sides of the neighboring taluka shapes did not match up exactly point-for-point. To fix this, I ran a line-merge tool on the broken pieces. It worked great and cleaned everything up into one single continuous line that is 384.7 kilometers long. It starts from the Akola, Amravati, and Wardha corner up in the north and goes all the way down to the Yavatmal and Chandrapur corner in the south.

I made a 25-kilometer buffer zone around that border line. Then I split it by division into a Berar side (7,387 square kilometers) and a Central Provinces side (7,987 square kilometers) to map out my near-boundary zone. Out of my 120 talukas, 32 fall inside this specific corridor area. They spread out across Amravati, Yavatmal, Wardha, Chandrapur, and a tiny piece of Nagpur district. This is the exact sample group I will use for my real border jump test in Phase 4 (I will not use the full 11-district dataset for that part).

Next, I generated a quick map to double-check my work. I overlaid the two divisions, the merged border line, and the buffer corridor right against the taluka grid. I just wanted to see with my own eyes that the line sits exactly where it belongs. It looks pretty even on both sides. There are no weird gaps or overlapping shapes at the border line.

What this is not yet. This is just a proxy using modern boundaries. I built the whole thing using present-day Census and POCRA taluka files. I still have not checked it against a real, old historical map from the Berar and Central Provinces times. The project roadmap notes say they are only "roughly the same" (they are definitely not proven to be exactly identical on paper yet). I still need to fix this. My best options for finding the old lines are the old Imperial Gazetteer maps (from the 1901 to 1931 books) or old Central Provinces and Berar atlases. The bad news is that I have not found a usable digital copy of either source yet.

Files I made in this step (`03_Data_Processed/Administrative/`): `divisions_dissolved.geojson` (2 big shapes), `division_boundary_line.geojson` (the 384.7 km line), `discontinuity_zone_25km.geojson` (the two-sided corridor), `boundary_verification_map.png`.

Things I still need to fix after this entry:

- I have not cross-checked the old history border against maps from that time yet (I still cannot find any good source).
- I still need to find the land use, economics, and FRA numbers (this is just carried over from my very first entry).
- My next step is Phase 3. I need to pull the actual results and variables (like cotton-dependency from NDVI, drought and LST numbers, and farmer stress facts) for the 32 talukas inside the buffer zone. Doing this will give my Phase 4 border jump test some actual numbers to run on.

Entry 3

I went looking for the climate heat data I talked about in Entry 1. Guess what? It was not where I expected it to be at all. The raw Landsat ST_B10 thermal maps, my two merged LST files (`vidarbha_lst_merge.tif` and `vidarbha_mergedLST.tif`), and a broken 8-byte file named `A1_LST_Celcius.tif` were all sitting inside the `Land_use` folder instead of the `Climate` folder. The actual Climate folder is completely empty. I am leaving this mess exactly as it is for now. Moving 3.5 gigabytes of heavy map data takes too much time. I am just writing it down here so I do not get confused later on. This also means I still have not found real LULC or land-cover data, even though that folder has files inside it.

The bug. Both merged heat maps were throwing back impossible temperatures—lowest was -123.15°C, which obviously makes zero sense for ground temperature on Earth. Found the stupid root cause for this glitch: Landsat Collection 2 Level-2 ST_B10 scenes use DN=0 as the fill/no-data value for cloud-masked and off-swath pixels (roughly a third of the scene). My old conversion script was blindly applying the Kelvin scale/offset formula directly to the raw DN data without masking out the zero values first. Of course, running DN=0 through that equation spits out exactly 149K, which is -124.15°C—the exact floor value both broken merges were stuck on.

I found the exact same coding mistake sitting in my plants and vegetation files too. The file `vegetation_mergedB4.tif` says it has 100% good pixels. That is totally impossible because the satellite pictures always have about 33% empty space. Those empty pixels were never blocked out. They are just sitting quietly in my merged map file as fake zero numbers.

The fix. I blocked out the DN=0 values as empty data *before* running the scale and offset math, not after. Then I re-ran the whole pipeline from the raw satellite files for both heat data (ST_B10 to Celsius) and light reflection data (SR_B4 and SR_B5 to true values). After fixing that, I calculated the greenness map using the standard formula: NDVI = (B5-B4)/(B5+B4). I made sure to cut out anything outside the real physical range of -1 to 1. Only a tiny group of pixels—exactly 4,750 out of 73.7 million—went totally wild because the bottom part of the math fraction came out close to zero near the map edges, so I just dropped them.

Scope of this pass. I did not want to fix and merge the full 12-scene map that covers all 11 districts right now. Instead, I just used the two main satellite scenes from WRS-2 path 144 (rows 045 and 046). Together, these two pictures cover 97.7% of the 25km border corridor area. This is the only zone where I am running my actual border tests anyway. There is a tiny uncovered slice (about 2.3% of the corridor space near the Amravati and Wardha border around latitude 21.30 to 21.46) where the two satellite photos do not quite touch. I am just marking this issue down here instead of wasting time downloading more heavy files, since it does not change the main data trend at all.

First read of the result. I looked at the 28 corridor talukas that have good pixel data. This includes 11 on the Berar Amravati Division side and 17 on the Central Provinces Nagpur Division side. My single June 2026 satellite snapshot shows that the Berar side runs hotter with a mean LST of 56.4°C against 54.6°C. It is also less green with a mean NDVI of 0.245 versus 0.292 right at the border area.

This matches what my main project idea guessed. But remember, this is just one single date before the monsoon rains hit. It is definitely not the full seasonal pattern that I actually need for the real cotton-dependency test. Also, my sample size is still very small with just 11 against 17 talukas. Because of this, I am treating these numbers as a first quick data point rather than a final result I can trust.

Files I made in this step:

- `03_Data_Processed/Climate/lst_fixed_300m_June2026.tif` — This is my corrected LST map. I changed it to 300 meters from the original 30 meters to save disk space, though I used the full-resolution version to calculate my numbers below.
- `03_Data_Processed/Climate/taluka_lst_ndvi_stats.csv` — A simple list showing the average heat and greenness numbers for all 66 talukas that have usable data inside my two satellite pictures.
- `03_Data_Processed/Vegetation/ndvi_fixed_300m_June2026.tif` — My corrected greenness NDVI map downsampled to 300 meters.
- `03_Data_Processed/Vegetation/corridor_taluka_lst_ndvi_stats.csv` — The exact same numbers list but cut down just for the 28 corridor talukas.
- `07_Outputs/lst_ndvi_corridor_comparison.png` — A final image file showing my greenness map, heat map, and simple box plots for both sides.

Things I still need to fix and track down:

- Those two old, glitchy merged maps (`vidarbha_lst_merge.tif` and `vidarbha_mergedLST.tif`) along with that broken 8-byte file are still sitting inside the `Land_use` folder. I will safely delete them once I completely verify the new fixed versions.
- I have not fixed or reprocessed the big 12-scene map that covers all 11 districts yet. I only worked on the 2-scene piece that matters for my border corridor. It might be worth fixing the rest later if I ever need a full district-level comparison.
- This is just a single snapshot from June before the monsoons started. My project roadmap actually asks for a multi-date greenness timeline across seasons (a single quick jump in green means a pure cotton

monoculture, while staggered shifts mean mixed crops). I need to get post-monsoon and pre-harvest satellite views before I can run that test properly.

- Land_use, Economics, and FRA files—I still have not found any good sources for these three.
- I have not even started on my farmer stress data (like MSP dependence numbers, crop yield changes, or population exposure calculations).
- **My ground-check idea to do later:** The cool thing is I am physically staying in Karanja Ghadge in the Wardha district right now. This specific taluka sits right inside my 25km border corridor on the old Central Provinces Malguzari side (my current June 2026 data says its average ground heat is 55.97°C and greenness is 0.264, but I only have 45.8% good pixels because it sits partly outside my current 2-scene satellite footprint). Here is my plan: I will go out and take real photos of 2 or 3 farm plots around Karanja Ghadge using GPS coordinates. I will note down the crop types, how thick the leaves look, and if the soil is wet. Then I will pull the exact NDVI and LST pixel numbers for those exact points to see if the satellite data matches what I can see with my own eyes. This will become a special ground-truth section on my final dashboard (it is a real way to check my satellite math, not just a random mention of my hometown). I am just waiting to take those field photos before I can move this task forward.

Entry 4

I am working on two things at the same time right now.

Seasonal NDVI scenes. The June map I made in Entry 3 only shows the time before seeds are planted. It is not the real cotton test. For that, I need a kharif peak date when cotton leaves are full after the monsoons, and a rabi peak date to see if farmers plant wheat or gram right after cotton. That second crop is the clear sign of mixed farming instead of growing just one thing. I cannot download new satellite images directly using my current setup. The network setup I have right here cannot reach a public STAC search API. So I wrote a Python script called `download_seasonal_scenes.py` and saved it right in the project root folder. It looks for the exact same WRS-2 path 144 and rows 045 and 046 footprint. It connects to Microsoft's Planetary Computer database because it is public, free, and needs no login. I set it to look for a post-monsoon block (September 15 to November 15, 2025) and a pre-harvest block (January 15 to March 15, 2026) to grab the clearest pictures with the absolute lowest cloud cover. I might need to run `pip install requests` if I do not have that package yet. Once the script finishes running, I will bring those pictures in and re-run my old zero-pixel fix and NDVI steps from Entry 3 across all three dates.

Farming stress proxy — first try using the data I already have. I did not need to search for new sources for this part. I built the whole thing using the APY 2024-25 Excel sheet and the Economic Survey 2025-26 District Indicators PDF file. Both were already sitting inside my Agriculture folder from before I changed my project topic.

I used the APY sheet to calculate each district's cotton area as a share of total farmed land. This total includes foodgrains, oilseeds, sugarcane, cotton, rabi jowar, and wheat because that is what this specific sheet actually covers. Then I opened the District Indicators PDF. I pulled out four main things for all 11 districts: total used farming land as a percentage of geographic area, food production per person, cropping intensity, and the average farm size. These numbers sat in a big table with merged headers from columns 25 to 31. I had to double-check this table carefully column-by-column against what I already knew about these districts before I trusted the layout. For example, my data showed Gadchiroli with only 14% farming land, while Bhandara and Gondia had over 500 kg of food production per person. This completely matches their real reputations as heavy forest zones and rice-bowl areas, so I knew I had the right columns lined up. I decided to leave out the

columns for MGNREGA work, malnutrition, and death rates (columns 52 to 59) on that same page. The PDF table extraction tool gave me a total mess there with bad alignment. I would much rather skip a variable completely than risk using a wrongly labeled number in my project.

What it actually shows, honestly. The cotton numbers are way more messy than the clean west-versus-east story I was expecting to find. Look at the data: Yavatmal sits at 51% and Wardha is right there at 49% (the funny part is, one was Berar and one was Central Provinces, yet they are almost tied for the highest cotton dependency in the area). Central Provinces districts like Chandrapur and Nagpur also run high at 38% each. Bhandara and Gondia sit at basically zero (0.15% and 0%), but that is simply because they are rice-growing areas, not just "not-cotton" spots. So cotton farming was never stuck only inside old Berar. The British forced it hardest there, but cotton growing spread into parts of the neighboring Central Provinces districts too. Wardha especially has a long cotton history completely separate from Berar's Ryotwari rules.

I found that net sown area is a much cleaner and more consistent line to look at. All five Amravati Division districts stay inside a tight 60% to 75% range. On the other hand, the six Nagpur Division districts are totally scattered from 14% in Gadchiroli (which is mostly heavy forest) up to 57% in Nagpur. This side averages around 41% and spreads out way more with a standard deviation of 15.4 against Berar's tiny 6.8. My data shows this is the real structural difference. Berar was completely cleared for farming. The Central Provinces stayed a mix of heavy crops and deep woods or water bodies. This makes total sense historically since the Ryotwari system forced almost all land into private farming, while Malguzari left large chunks under big estate or forest control that people never fully turned into crop fields.

Files I made in this step:

- `download_seasonal_scenes.py` (sitting in the project root folder) — This is the script I wrote to grab the two extra Landsat dates later.
- `03_Data_Processed/Agriculture/vidarbha_agrarian_indicators.csv` — My main table containing cotton share, net sown percentage, food production per person, cropping intensity, and average farm sizes for each district with division tags.
- `07_Outputs/agrarian_indicators_comparison.png` — My final four-panel chart comparing these indicators visually.

Things I still need to fix and clear up:

- I need to actually trigger that `download_seasonal_scenes.py` file. I have to get those post-monsoon and pre-harvest satellite maps back to my disk so I can build the real cotton-signature timeline test.
- I skipped the MGNREGA work days and child malnutrition numbers from the District Indicators PDF (columns 52 to 59). The bad column layout made me nervous, but I might try a second extraction pass later if I realize these numbers are useful for my final writing.
- Land_use, FRA, and separate Economics data are still totally empty (carried over again).
- I have not touched the village-level census population numbers that I already downloaded. My stress proxy so far only uses basic farming data. I know a population version showing the exact share of workers who are farm laborers versus landowners will make my project way stronger. That is my next logical step.

Entry 5

While my script `download_seasonal_scenes.py` was running on my machine, I went ahead and opened up the village population data. I downloaded this census data a while back but never actually clicked open the files until today. I focused only on the five districts that matter for my border corridor: Amravati, Yavatmal, Wardha, Chandrapur, and Nagpur. That means dealing with 9,161 villages and over 600 data details for each one of them.

I grouped the village numbers up to the taluka level to figure out two main things. First, I calculated the ratio of farm laborers to landowning farmers (I mixed the main and marginal workers together since this shows real landlessness and stress when people have to work someone else's land for cash). Second, I pulled the literacy percentages.

I tried to grab irrigation numbers too but I had to throw that data out completely. The file columns named `area_irri` and `net_area` showed irrigation rates between 58% and 92% for Yavatmal villages. This makes no sense. It completely goes against the well-known fact that Vidarbha's cotton fields depend almost entirely on rain (real numbers usually sit way lower around 15% to 20%). So tried dividing the numbers by the total village area as a backup test, but the results were still way too high to be real. Whatever these column names mean, they do not show simple watering data, so I chose to leave a blank space instead of using numbers that I know are wrong. I will try to find a data dictionary guide for this map shapefile later to see what is actually going on inside those columns.

What held up. I looked at the agricultural worker numbers for the 32 corridor talukas where my data matched up. This includes 15 on the Berar side and 17 on the Central Provinces side. The numbers show a big gap: 2.90 on the Berar side compared to 1.96 on the Central Provinces side. This means for every single landowning farmer in the Berar talukas, you have nearly three hired laborers, while the other side averages just about two. This is a real, steady pattern. It is not just a statistical mistake caused by one or two strange talukas. It perfectly fits the old history books. The Ryotwari system gave normal peasants actual papers for their land, but forcing them to grow only cotton created a dangerous cash economy tied to market prices. Over generations, this specific cash crop setup makes people lose their farms and forces a massive class of people to become landless workers. This fact is the main point of all the real books written about farming trouble in Vidarbha (it is definitely not my own random idea, as this is the standard reason why the worst farmer suicide crises always concentrate in this exact western belt). I also checked literacy, but the numbers were super close at 75.1% against 72.5%, so it is not a big difference on its own.

Files I made in this step:

- `03_Data_Processed/Population/taluka_population_agri_stats.csv` — My full list of grouped taluka numbers showing total population, houses, the worker breakdown, and basic reading skills across the 5 districts.
- `03_Data_Processed/Population/corridor_taluka_population_stats.csv` — The exact same table but chopped down just for the 32 corridor talukas with their proper division names.
- `07_Outputs/population_agrarian_comparison.png` — A final image file where I made simple box plots to compare the worker ratio and literacy numbers across the old border.

Things I still need to fix and check:

- The watering and irrigation data from this file is totally broken. It is just not trustworthy as it stands now. I need to take a second look or find a completely different file. Data directly from the state agriculture department would be much safer than trying to guess what this map file means.

- I only worked on 5 out of the 11 districts (just the ones touching my border corridor). The other 6 districts (Akola, Buldhana, Washim, Bhandara, Gondia, and Gadchiroli) are completely untouched. I made this small-scale choice on purpose, matching my earlier satellite heat maps.
- I noticed this census data also has village land fields for forests, empty land, and rocks. I could use these later to partially fill my missing land use category. It is definitely worth coming back to.
- I am still waiting for my script `download_seasonal_scenes.py` to finish running on my computer. Once it stops, I can finally start my real multi-date cotton timeline math.

Entry 6

I found out that the same census map files from Entry 5 actually have a full land list for each village too. This includes forests, non-farming space, rocks, empty fields, tree areas, and used crop areas. The numbers are written in hectares, and they match the total village sizes perfectly. I checked this math myself. All nine sections add up to nearly 100% of the total size for every single village. So, unlike those broken watering numbers from my last entry, I completely trust this data. This fixes my missing land use category that was empty since Entry 1. It is not satellite land cover, but a real independent list from the census. This is actually much better for my project because the dividing work for forests and fields is already done by hand. I do not have to write or train a custom coding tool for it.

I checked these numbers very hard before trusting them. Chikhaldara is a hilly tiger-reserve area in the Amravati district, and my code showed it has 30.9% forest. Jiwati in Chandrapur is a very famous woody area, and it came out at 36.4% forest. Both numbers completely match what people know about these places, which is what proved to me that these columns are real.

Border corridor check for my 32 talukas. My data shows that used farming land is 68.9% on the Berar side compared to 61.5% on the Central Provinces side. This goes in the exact same direction as the district numbers I found in my Entry 4 PDF file. But the gap is much narrower here. Why? The simple reason is that my buffer zone only touches the heavily farmed parts of the Central Provinces (like Wardha, Nagpur, and Chandrapur). It misses the massive, deep jungle tracts of Gadchiroli that pulled the whole district average down earlier. Forest numbers look almost exactly the same right at the border (12.5% vs 12.6%) for this exact same reason. This corridor is just a shared farming zone, not where the real heavy woods of the Central Provinces sit. But I did stumble on one new finding here. Empty or fallow land is nearly double on the Central Provinces side of the border area (5.8% against Berar's 3.3%). That means more land is sitting completely idle over there. I am not making a big deal out of this point yet. Normal crop rotation can easily explain it just as well as financial trouble. Still, I will definitely add this fact to my final report alongside my numbers on cotton shares and farm laborers.

Files I made in this step:

- `03_Data_Processed/Land_use/taluka_landuse_stats.csv` — My full list of land use columns for all matching talukas across the 5 districts.
- `03_Data_Processed/Land_use/corridor_taluka_landuse_stats.csv` — The same data list but cut down just for my 32 corridor talukas with their division names.
- `07_Outputs/landuse_corridor_comparison.png` — A final image file showing my comparison plots for farming land, empty fields, and forests by division.

Things I still need to finish and track:

- This land data comes purely from the village census books, not from a direct satellite scan. It would still be awesome to grab an actual LULC map file (like from ISRO Bhuvan or somewhere similar) later on so I can do a proper tech check. But the main thing is my land use gap is finally filled.
- I stuck to the same 5-district limit here. It is just the corridor area, exactly like what I chose to do in Entry 3 and Entry 5. The other 6 districts are still completely blank and untouched.
- My script `download_seasonal_scenes.py` finally finished downloading all the files while I was finishing up this task. My next plan is to bring those satellite pictures into my workspace so I can run my actual cotton-timeline greenness math.

Entry 7

This is the real Phase 3 result variable. I am finally replacing that temporary June data from Entry 3 with actual seasonal mapping. I have three complete dates on disk now: June (before planting), November 2025 (the high point of the kharif monsoon), and March 2026 (the rabi pre-harvest time). I ran my old zero-pixel correction code on these two new satellite images. Then I calculated the actual seasonal greenness drop for each taluka by checking how much the NDVI value falls between the November high point and the March winter period. If the drop is huge, it means the fields go completely bare after the monsoon crop is harvested (which is the classic signature of a single cotton crop). A smaller drop means farmers are planting a winter crop like wheat or gram right after the first harvest. That keeps the land green through the cold months, showing a mixed farming setup.

My results across the full footprint (68 talukas across both satellite rows). The Berar Amravati Division side shows an average greenness drop of 0.325 NDVI points. The Central Provinces Nagpur Division side sits at 0.255. That is a visibly smaller drop, and it spreads out way more with a standard deviation of 0.062 against Berar's tight 0.041. This tells me that the winter crop pattern is very real but uneven on the Central Provinces side instead of being the exact same everywhere. The crazy thing is both sides hit almost the exact same high greenness score during the monsoon (0.66 against 0.64). The actual difference is not about how green things get when it rains. It is all about what happens after the rains stop. That is the exact trend I wanted to find from the very start. I am looking at a single-crop setup that sits empty once cotton is picked, compared to a mixed-farming setup that keeps parts of the land working through the dry winter season.

At the border corridor specifically, this gap almost disappears. The numbers sit at 0.310 versus 0.299 across my 29 corridor talukas. This matches the exact same trend I saw with the cotton shares back in Entry 4. The Central Provinces talukas right inside my buffer zone (like Wardha, Nagpur, and Chandrapur) grow a ton of cotton themselves. So right at the border line, the two sides actually look very similar compared to the wide district averages. The trend is real across the whole region, but it does not make a sharp jump at the old historic line. It behaves like a slow trend that gets stronger as you move further away from the border in both directions. It is definitely not a clean, sharp drop. I need to be completely honest about this fact before I start Phase 4. A spatial jump test run only on the tight corridor area might show a much weaker link than the wide regional numbers suggest. That is a real, honest finding about how this old border behaves on a map (it is definitely not a mistake or a failure of my research method).

Files I made in this step:

- `03_Data_Processed/Vegetation/ndvi_postmonsoon_300m.tif` and `ndvi_preharvest_300m.tif` — These are my two new fixed greenness maps downsampled to 300 meters to save disk space.

- `03_Data_Processed/Vegetation/taluka_seasonal_ndvi_signature.csv` — My main list showing all three dates and the greenness drop for all 68 matching talukas.
- `03_Data_Processed/Vegetation/corridor_taluka_seasonal_ndvi_signature.csv` — The exact same table list but cut down just for my 29 corridor talukas.
- `07_Outputs/seasonal_ndvi_signature.png` — My final image file showing the seasonal timeline chart and the greenness drop box plots.

Things I still need to finish and track down:

- This data only covers one single farming year (the kharif 2025 and rabi 2025-26 seasons). I grabbed these dates because they sit a year away from my main June 2026 starting point. I should probably get a whole second year of satellite maps later. That will prove my numbers do not just show one weird or unusual season.
- I have not started Phase 4 yet. That is the actual math part where I run the spatial-discontinuity statistical tests while blocking out the effects of distance, height, and rain. This entry just gave me the final greenness drop numbers that the main test needs to run on.
- The land use and FRA sections are still completely open from my earlier entries. Also, my field-check idea for Karanja Ghadge is still sitting idle. I am just waiting on the actual farm photos to come in before I can finish that part.

Entry 8

I finally pulled in the WorldClim rain data that I already downloaded a while back. It has been sitting completely untouched on my laptop since my original data pull. I sampled all 12 months at the center point of every single taluka. Then I calculated the total annual rain, the percentage that falls during the monsoon months (June to September), and the month-to-month changes.

A super important note about the setup before looking at numbers: this is a rainfall *climatology* map (meaning long-term monthly averages over decades, not the actual rain from one specific year). Also, the block size is very rough. WorldClim's 10 arc-minute grid means each pixel is about 18.5 kilometers wide. That is actually bigger than most single talukas, so neighboring talukas often end up sharing almost the exact same rain numbers. It works totally fine to show the big regional trend, but it is just not sharp enough for tiny taluka-by-taluka precision.

What the data shows: the Central Provinces side gets way more rain than the Berar side. The wide district numbers show 1,249 mm per year against 903 mm per year. This shrinks down to 1,109 mm versus 1,005 mm right at the border corridor area. The monsoon concentration is almost identical on both sides (around 88% of the yearly rain falls between June and September either way). So the real difference is just in the total amount of water, not the timing or seasonal shape.

This rain gap is definitely NOT a finding about the old historic border line. It is just a natural west-to-east rain trend across Vidarbha that has nothing to do with old Ryotwari or Malguzari rules (nature made this trend way before colonialism existed). I am writing this down here because my project roadmap specifically said I must block this out in my Phase 4 math. If Berar is historically a cotton zone AND naturally drier, any result saying "Berar looks more drought-stressed" might just be because it gets less rain, not because of old land laws. My Phase 4 models need to clear this out completely before I can trust any border effect.

Files I made in this step:

- `03_Data_Processed/Hydrology/taluka_precipitation_stats.csv` — A list of monthly and yearly rain numbers for all 120 talukas.
- `03_Data_Processed/Hydrology/corridor_taluka_precipitation_stats.csv` — The exact same rain list but chopped down just for my corridor talukas.
- `07_Outputs/precipitation_corridor_comparison.png` — Simple box plots comparing the total yearly rain and monsoon shares across the two sides.

That finally closes out Phase 3. I have my cotton-signature NDVI numbers from Entry 7, my farmer stress and land use data from Entries 4 to 6, and now my climate control variables are ready. I am moving straight to Phase 4 next. It is finally time to run the actual spatial border jump tests.

Entry 9

This is the real test that my whole project has been building toward from day one. I want to know if each result jumps suddenly right at the old historic border line once I block out the effects of distance, rain, and height. Or is everything I wrote down in Entries 3 to 8 just a broad regional trend that happens to look different on each side of the line?

Getting my dataset together. I needed elevation numbers as a control variable, so I grabbed 9 SRTM terrain tiles that cover my corridor box. Then I sampled the height at the center point of all 32 corridor talukas. The numbers came out between 194 and 497 meters, which makes total sense for this part of the Deccan plateau edge. Everything else in this big table comes directly from my earlier entries. I pulled the `side` tag (where 1 means Berar and 0 means Central Provinces) and the `signed_dist_km` numbers (which shows how many kilometers each taluka center is from the merged border line, calculated in UTM 44N and written as positive on the Berar side) from my Entry 2 boundary files. I got the three-date NDVI and the monsoon-to-winter greenness drop from Entry 7. The ground heat numbers come from Entry 3. I took the worker-to-farmer ratio and literacy rates from Entry 5. My land use percentages come from Entry 6, the total yearly rain data comes from Entry 8, and the cotton area shares come from Entry 4. This leaves me with exactly 32 corridor talukas, with one neat row of facts for each one.

The model. I used a standard regression-discontinuity setup. The math formula is `outcome ~ side + signed_dist_km + side×signed_dist_km + rainfall + elevation`, and I ran it using OLS. The `side` variable is the most important part because it catches the sudden level jump right at the border line. That is the actual question I am asking here. The interaction term shows if the *slope* or trend with distance is different on each side of the border (meaning, does the effect grow stronger or weaker as you travel away from the line, and does that change across the two sides?). I ran this code for five results: the kharif-to-rabi NDVI greenness drop (I ran this both with and without my control variables to see what changes), the farm laborer ratio, the used crop land percentage, and the June ground heat numbers.

Being blunt about what came back: at n=28–32, most of the side coefficients don't clear statistical significance. NDVI decline side p=0.458; agricultural-laborer ratio side p=0.960; net cultivated area p=0.577; land-surface temperature side p=0.792. None of these show a clean level jump right at the boundary once the geographic controls are turned on. No point dressing that up—a corridor-only sample this small has limited power to detect anything short of a massive, sharp effect. Plus, Entry 7 already flagged that the cotton-signature gap itself nearly vanishes at the immediate corridor even before any regression. This math just confirms that.

The one result that did show real significance. The `side_x_dist` interaction number for the farm worker ratio came back with a coefficient of 0.0374 and a p-value of 0.036. It is not a sudden jump at the line. Instead,

it is just a difference in the *slope*. As you travel away from the old border, the laborer ratio changes at a completely different speed on the Berar side compared to the Central Provinces side. I am reading this very carefully rather than screaming that I found a massive border jump. It basically tells me that the link between distance from the old line and the worker setup is not flat. It has a different shape on each side of the border (which is cool on its own). But this is just one single winning number out of twenty across five different math models. With my small sample size, I definitely want to double-check this before using it as a major fact in my final writing.

What did stay significant every single time: my control variables. Rain and ground height show up as real drivers in almost all my models. Look at the numbers: rain hit $p=0.015$ and height hit $p=0.002$ in the greenness drop model. Rain was super strong at $p<0.001$ in the farm laborer model. Then both rain ($p=0.046$) and height ($p=0.020$) showed up big in the used farming land percentage model. This is exactly what I flagged back in Entry 8 before starting this step. The actual nature and geography of Vidarbha (the Central Provinces side is simply wetter and lower, while the Berar side is drier and higher) does the real work here. It changes these results completely on its own, separate from any old colonial line. I am really glad this is coming through safely as a basic control variable instead of getting mixed up with a fake border effect. That is the exact reason why I built Entry 8 before running these tests.

My honest summary of the whole thing. Inside my tight 25km buffer corridor and with this small group of talukas, I am simply not finding a sharp mathematical jump right at the old historic line. Instead, I am seeing a real regional trend that I wrote down across Entries 4 to 8. This trend slowly fades out as you get close to the actual border line. I also found one slope difference that is worth pointing out instead of a sudden level jump. This is a perfectly fine and real finding. I will definitely not hide or bury this zero result. It simply tells me that the history link I am looking for does not look like a hard, sharp edge right at the old line. It shows up as a much wider east-to-west pattern across Vidarbha. My small corridor test just does not have enough power to pin that down properly. I want to take two main ideas forward into Phase 5. First, I want to see if using a wider corridor or testing the full 68-taluka footprint gives my math models more power without losing the border logic. Second, I need to remember that finding a zero result at the line but a trend across the whole region is a real part of the story, not some annoying mistake.

Files I made in this step:

- `03_Data_Processed/Phase4/phase4_master_dataset.csv` — My big master math table with 32 rows and 19 columns for the tests.
- `03_Data_Processed/Terrain/corridor_taluka_elevation.csv` — My list of corridor taluka heights that I got from the SRTM tiles.
- `07_Outputs/phase4_discontinuity_plots.png` — My final image file showing the scatter plots and fitted trend lines for my four main results.

Things I still need to finish and track:

- My small sample size is the biggest problem here (it sits between 28 and 32 talukas). It is worth testing if the full 68-taluka footprint gives me better mathematical power. I know that moves away from a tight border test, but it is worth a shot.
- I need to double-check that one winning interaction result for the farm laborer slope. It needs a serious robustness check before I write it down in my final paper (right now it is just a single number, not a proven pattern yet).
- My field check at Karanja Ghadge is still stuck. I am just waiting on the actual farm photos from Sawardoh village to come in.

- The old map cross-check, the FRA data list, and the second year of greenness data are all still completely open from my earlier log entries.

Entry 10

I have three things left over from Entry 9, and I am handling them one by one.

Farm splitting check. I mapped the `avg_holding_ha` column from my Entry 4 district table onto the 32 corridor talukas to check the setup. The Berar corridor talukas average 1.81 hectares per farm, while the Central Provinces side averages 1.90 hectares. So the Central Provinces side actually has slightly *bigger* farms, not smaller ones. This goes completely against the simple guess that the Ryotwari system always leaves behind smaller, more broken-up pieces of land today. Both numbers are actually super close. They stay within a tiny quarter-hectare gap with a very similar spread, so I see no real difference here either way. This is just a flat zero result across the line.

Double-checking the Entry 9 slope result. I ran a leave-one-out jackknife test on the one significant result I found in Entry 9. This was the `side_x_dist` interaction for the farm worker ratio. I re-ran my math model 32 times, dropping a different corridor taluka each time to see what happens to that specific number. It stayed positive and sat inside a tight band between 0.026 and 0.044 every single time. That is a great sign — no single taluka is secretly faking or driving the whole mathematical trend on its own.

But the p-value was a lot more jumpy. It stays below 0.05 in 26 out of the 32 runs, but it crosses back over the line (climbing as high as 0.138) when I leave out a few specific talukas like Amravati taluka itself, Seloo, Samudrapur, Narkhed, Korpana, or Yavatmal taluka. So the basic direction and rough size of the effect look steady, but the statistical value sits right on the edge. A few single talukas are doing most of the heavy lifting to push the numbers under the $p<0.05$ line. I am calling this result a good hint rather than a solid, bulletproof fact. I will definitely not oversell it or throw it away completely.

Rerunning the test on the full footprint. This is the much bigger task. My old Entry 9 math only used the 32 corridor talukas, and I already noted that this sample was simply too weak. Luckily, both greenness (NDVI) and ground heat (LST) have full data for the whole satellite scene sitting in Entries 3 and 7 (that gives me 68 and 66 talukas from the complete path-144 view, not just the small 25-kilometer buffer zone). So I re-ran those two tests using this much larger group. I had to download two extra SRTM tiles named n20_e080 and n21_e080 to map out the height data for this wider area. I merged and sampled them cleanly. All 68 talukas got a correct height number this time.

My other data points like population, land use, and the farm worker ratio do not have this choice. That specific census data only covers 5 districts of Vidarbha on my disk. So those files must stay restricted to the smaller corridor for now. Expanding them would mean searching for and processing village maps for three more entire districts. That is a massive, separate job, not a quick thing I can just add on today.

I must point out one big issue before reading the results: the full satellite footprint is totally uneven and lopsided. It has 49 Central Provinces talukas but only 19 Berar talukas. This happens simply because the satellite camera view captures way more territory on the Central Provinces side. I am keeping this bad match in mind when I read my final numbers.

What changed with the bigger sample. The sudden jump number at the border (`side`) is still completely useless in both models. For the greenness drop, the p-value came out to 0.821, and for ground heat, it sat at

0.512. This perfectly matches what I guessed in Entry 9. There is just no sharp jump right at the old line. But the trend interaction number, which only worked for one single variable in my smaller test, is now significant for *both* satellite outcomes across the wider map footprint. Look at the data: the greenness drop side-by-distance p-value is 0.035, and the ground heat side-by-distance p-value is 0.010. Also, `signed_dist_km` by itself became super strong in both models with a p-value under 0.001.

This is a solid point. There is a real, well-powered trend that changes as you travel away from the old border line. It just takes a completely different shape or slope on each side. This is a much cleaner and steadier version of what I already suspected from my tight corridor test. It is not a sudden jump at the line. It is a genuine difference in how these trends behave with distance depending on which side you stand on. Finding the exact same shape of results across two completely different physical measurements (and with real statistical backup this time) is the absolute best proof I have for this story.

Files I made in this step:

- `03_Data_Processed/Agriculture/phase4_master_dataset.csv` — I updated this file by pasting in the new `avg_holding_ha` column.
- `03_Data_Processed/Phase4/phase4_robustness_jackknife.csv` — My master list showing all 32 drop-one-out coefficients and p-values from the test.
- `03_Data_Processed/Phase4/phase4_fullfootprint_dataset.csv` — My big 68-taluka wide math table for the regressions (this only has NDVI and LST data because of the missing census files I noted before).
- `07_Outputs/phase4_fullfootprint_plots.png` — My final image file showing the greenness drop and ground heat trends with distance from the border line.

Things I still need to finish and track:

- My farm laborer numbers, used crop land percentages, and land use columns are still stuck inside the smaller 32-taluka corridor. Extending these to the wider map footprint means I must search for and process population and land use census records for Bhandara, Gondia, and Gadchiroli districts (I have not sourced these extra files yet).
- The slope trend I found in Entry 9 looks much stronger now. The exact same slope-difference pattern shows up completely independently across my larger greenness and ground heat data. Having three separate things pointing the exact same way shows a real trend, not final proof, but it is easily the strongest sign I have found so far that my project is about a slow regional gradient rather than a sharp border jump.
- My field check for Karanja Ghadge is still completely stuck while I wait for the farm photos from Sawardoh village.
- The old history map cross-check, the messy FRA data list, and my second year of greenness math are all still completely open from earlier log entries.

Entry 11

I found out that I did not actually need to search for new data for Bhandara, Gondia, and Gadchiroli. The raw census map zip files (`bhandara.zip`, `gondia.zip`, and `gadchiroli.zip`) were already sitting quietly in my Population folder alongside the other 8 district files. I had just left them unzipped and untouched since downloading them. So I ran them through the exact same village-to-taluka grouping step that I used back in Entry 5 and 6. I added up all the worker and land use columns by district and taluka names, and then I

calculated the ratios again. All 27 talukas across these three extra districts came out completely clean. Their land use category numbers add up to exactly 100% of the total size in hectares for every single one of them. Also, the spelling of the taluka names matched my administrative map layer perfectly (thankfully, there was no annoying "Gondia versus Gondiya" text mismatch to fix this time).

With these new numbers in hand, I rebuilt my full-footprint Phase 4 master table. This time, I could include all four main result variables (greenness decline, ground heat, farm worker ratio, and used crop area percentage) instead of just the two satellite options I had in Entry 10.

The greenness drop and ground heat numbers came back basically identical to Entry 10. There is still no sudden level jump at the old border line. The p-values hit p=0.821 for the NDVI drop and p=0.512 for the LST heat maps. But the trend interaction term was still significant for both of them with a greenness side-by-distance p-value of 0.035 and a heat side-by-distance p-value of 0.010. That specific part of my project story holds up great.

The farm worker ratio did something totally different, and I need to be completely honest about it. Back when I checked just the tight corridor area in Entry 9, this specific number showed a real trend difference in the *slope* with a p-value of 0.036, but it had no sudden level jump. Now that I am using the full footprint data with a sample size of 67 talukas (this includes Bhandara and Gondia now, though Gadchiroli's talukas sit outside my actual satellite picture footprint so they did not make it into this specific test, even though I have their processed facts safe on disk for later use if I widen the map), the whole result flips. Now I found a significant level jump right at the border line under the `side` term with a coefficient of 0.622 and a p-value of 0.014. Meanwhile, the slope trend number completely loses its importance at p=0.579.

This is not a real mistake or a direct contradiction in my math. But it definitely needs a very honest explanation instead of me just cherry-picking whichever result looks prettier for my paper. I went and checked the actual distance ranges on both sides of this larger sample to figure out why this happened. The Berar side barely stretches past 20 kilometers from the actual border line here (this happens because Amravati is the only district from the Berar division that actually fits inside this specific satellite view). The deeper interior parts of Berar like Akola, Buldhana, and Washim were simply never inside this particular path-144 camera path to begin with.

On the other hand, the Central Provinces side now reaches way out past 160 kilometers because Bhandara and Gondia sit that far to the east. So I am no longer comparing two neat parallel trends here. It is just a short, border-hugging Berar piece up against a massive, far-reaching Central Provinces line. Reading this as a level jump near the border makes a lot more sense than guessing a different slope across the whole area, since only one side actually has enough map range to show a real slope line in the first place. I trust my smaller corridor-only trend results from Entry 9 much more when talking about what happens right at the line. I am treating this full footprint jump as a hint about a much bigger regional gap across the two divisions rather than a double test of the exact same thing.

Used farming land percentages stayed flat either way. The sudden level jump variable completely failed to show anything important across both tests. My numbers came out at p=0.121 for the wide footprint versus p=0.577 for the tight corridor. It is a very clean zero result across both ways of testing. I see no need to split or force a hidden story there.

Files I made in this step:

- `03_Data_Processed/Population/taluka_population_agri_stats.csv` and `03_Data_Processed/Land_use/taluka_landuse_stats.csv` — I updated both tables to cover 8 out of the 11 districts now. I successfully added Bhandara, Gondia, and Gadchiroli into the files alongside the original 5 districts.
- `03_Data_Processed/Phase4/phase4_fullfootprint_dataset.csv` — I updated this master math table to include all 4 result variables across 67 to 68 talukas depending on the test.
- `07_Outputs/phase4_fullfootprint_all4_plots.png` — My final image file plotting all four large footprint results against distance from the border line.

Things I still need to finish and check:

- I processed and saved Gadchiroli's population and land use numbers, but the district has no matching satellite map coverage here. These files are only useful if I choose to download a much wider Landsat map area later on.
- The farm laborer numbers now give me two completely different, totally fair reads depending on which sample group I run the math on. I must point this out very clearly in my final paper instead of quietly hiding one result to make my life easy.
- My field check at Karanja Ghadge is still stuck while I wait for the farm photos from Sawardoh village. The old history map cross-check, the messy FRA data list, and my second year of greenness math are all still completely open from earlier log entries.

Entry 12

I am handling two of the remaining open things together in this session.

FRA data list. I marked this all the way back in Entry 1 as a total mess with no clean source found yet. I knew it would need manual typing from old PDF files, so I went looking properly this time instead of leaving it blank. Here is the honest truth before looking at any numbers: there is absolutely no clean taluka list, or even steady district list, for Forest Rights Act data across Vidarbha online. I searched everywhere. This is not a case of me being lazy or not searching hard enough. What actually exists is broken across a few different files at completely different scales:

- I found Maharashtra state totals from two separate years. First, from the state Tribal Development Department's 2020-21 report up to March 2021: they got 379,881 personal rights claims, finished 95.84% of them, and gave out 1.31 lakh acres. For group community rights, they got 11,985 claims, finished 81.59%, and gave out 28.28 lakh acres. Second, a 2024 committee report showed newer numbers up to June 2023. They added 35,654 new personal claims and 90,626 new personal titles since 2017. But they only gave 1,290 new community group titles in that same time (30% of claims are still totally stuck statewide and 18% were completely thrown out).
- I found one genuinely useful district number for Gadchiroli specifically from that same 2024 report using older 2016 facts. It shows 1,355 community group titles covering 4,34,181 hectares against a target of 5,57,137 hectares. That means 77.9% was successfully cleared. The report explicitly calls it a "high performing" area, comparing it directly to Nashik district on the other side of the state which sits at a tiny 5%. That is a real, sourced, specific fact on disk (but it is just for one single district and almost ten years old now).
- I found a wide regional number for Vidarbha that is not split by district. It shows roughly 6,500 villages and 7,94,118 hectares given out under group community forest rights since 2013 (I read this in an EPW

magazine paper mentioned inside a Scroll.in news report talking about Gondia's forest work).
- I also found a few single village examples with real numbers, but there is no way for me to add them up into a full district total. For example, Menda-Lekha village in Gadchiroli got 1,800 hectares back in 2009 and made 75 lakh rupees from selling bamboo by 2012. Dhamditola and Paulzhola villages in Gondia secured 290 and over 600 hectares in 2013.

I also discovered that the Ministry of Tribal Affairs runs a live public web dashboard at forestrights.nic.in/mpr/common/dashboard. In theory, this website lets you click and see state or district breakdowns. But it is an interactive portal, not a file I can easily download, and I simply cannot reach it automatically from my code workspace here. If I ever absolutely need district-by-district numbers for Gadchiroli, Chandrapur, Gondia, Bhandara, Yavatmal, Wardha, Nagpur, and Amravati later on, that website is the only place to get them. Someone would have to click through it by hand, one single district at a time.

Because of all this data mess, I am treating the FRA numbers as a background history story for my final writing, not as a main math variable for my Phase 4 models. There is simply no steady or equal measurement across enough districts to paste it into my taluka table the way I did with my greenness or rain data. It works fine as a background history fact. Gadchiroli's high land recognition rate is a very cool number to talk about when I describe the forest rules of the region. But it simply cannot do any statistical heavy lifting in my models the way my satellite and census facts can because of the missing district slots.

Second-year NDVI check. I wrote a new script called `download_seasonal_scenes_year2.py` to run locally on my machine. It follows the exact same style as my first seasonal download script and grabs the same WRS-2 map box (path 144, rows 045 and 046). The big change is that I pushed the timeline back by an entire year. Now it looks for a monsoon kharif high point between September and November 2024, and a winter rabi peak between January and March 2025. This is the exact double-check I talked about in Entry 7. Back then, I found that the greenness drop was 0.325 in Berar versus 0.255 in the Central Provinces, but I only measured that on one single farming cycle. If I see the exact same gap (or something very close to it) show up completely on its own in the older 2024 data, that is real evidence I found a steady, real farming pattern — not just something caused by one weird or unusual season. I am handing this script over now. It needs to run on my local laptop and pull from Microsoft's Planetary Computer catalog exactly like the first file did.

Files I made in this step:

- `03_Data_Processed/Economics/fra_compilation_vidarbha.csv` — I put everything I found above into this file. It has one row per number and labels each fact by its scale (state, district, or village), its unit, the date, and the source web link. I built this purely as a typed-out summary because the raw files are a broken mess, not a clean data table.
- `download_seasonal_scenes_year2.py` (sitting right in the project root folder) — This is ready for me to hit run on my machine, matching the exact code pattern as the first download script.

Things I still need to finish and track down:

- For the FRA data, only Gadchiroli has a real, solid number. The other 7 districts have absolutely no district-level numbers that I could find online. So this has to stay as a background text point for my final writing, not a main math variable. The only fix is if someone clicks through the MoTA dashboard by hand later on.
- For my second year of greenness data, I am currently waiting for my script to finish running. Once those 2024 satellite maps land on my computer, I will re-run my old zero-pixel correction and NDVI steps on them. Then I can compare that winter greenness drop directly against my 2025 data from Entry 7.

- My field check for Karanja Ghadge and Sawardoh photos is sitting idle. I am holding off on this part completely until I get told to pick it back up.
- My old history map check is still totally open. I still cannot find any old period maps to check against my modern lines.

Entry 13

My project roadmap always told me to find a digital 1901 to 1931 Imperial Gazetteer map and paste it over my modern division lines. I searched everywhere. I never found a digital old map file like that. It simply does not seem to exist anywhere in a way I can actually open.

But what I found instead does the exact same job in a different way. I dug up the actual written lists of districts that were inside the old Berar Division and the Central Provinces Nagpur Division from 1903 to 1956. This lets me check if the exact same land sits on each side of my modern map line district-by-district. I do not even need a pixel-level digital map overlay anymore.

Old Berar Division history check (1903-1956). The old papers list these main places: Amraoti (Amravati), Akola, Ellichpur, Buldhana, Basim, and Wun. I cross-checked these names against what we call these places today. Ellichpur is now just a single taluka inside the Amravati district, not its own separate district. It got mixed in very early on. Wun was renamed Yavatmal. Basim was renamed Washim (it was actually folded into the Akola district for a long time before they remade it as a separate district on July 1, 1998).

The final result of my check is very clear: **the old historical Berar map fits district-for-district onto my modern Amravati Division area (Akola, Amravati, Buldhana, Washim, and Yavatmal).** It is the exact same 5-district territory. People just changed the names and split the land up internally over the last 120 years, but nobody ever redrew the main outer border line. This is a very clean and reassuring match for my whole project premise. It means I can safely use my modern division merge tool as a stand-in for the old historical Berar line.

Old Central Provinces Nagpur Division history check. The old records list these areas: Nagpur, Bhandara, Chanda (which people renamed Chandrapur), Wardha, and one real messy problem—**Balaghat**. Gondia (which split from Bhandara in 1999) and Gadchiroli (which split from Chandrapur back on August 26, 1982) are just later internal cuts of that exact same old territory. They are not new land pieces, so they match up nicely. But Balaghat is a different story. On November 1, 1956, when the government redrew state borders based on languages, Balaghat got moved into Madhya Pradesh's Jabalpur Division. The rest of the old Nagpur Division spoke Marathi (Nagpur, Bhandara, Wardha, and Chandrapur). So those parts went to Bombay State and eventually became part of Maharashtra in 1960.

What this actually means for my project. Balaghat was never sitting inside my files to begin with. It is not a Maharashtra district today, so it was never going to be part of my 11 Vidarbha districts anyway. Absolutely nothing about my Phase 2 border mapping needs fixing. But I want to be super exact when I write down my final paper later on. My Central Provinces side in this project is specifically just the Maharashtra-Vidarbha piece of the old CP Nagpur Division. This makes it slightly smaller than the full historical area from 120 years ago. The piece that is missing fell out because of a 1956 language border change, which has nothing to do with the farming land laws I am actually testing. I am really glad to have this written down as a clean, simple note rather than a weird blank space that I cannot explain.

The bottom line of my check. The modern Amravati and Nagpur division border line that I built back in Entry 2 is exactly the same line as the old historical Berar and Central Provinces border. I proved this

completely by tracking down the written office history of every single district instead of laying an old map image over my modern files. It would still be super cool to find a real, old digital map from the 1900s to use as a basic picture in my final paper later on. But the main question my roadmap was actually asking—whether my modern border map sits exactly where the old historical one used to be—now has a very well-sourced yes.

Files I made in this step:

- I did not save any new files inside `03_Data_Processed/`. This pass was just me doing reading and documentation checks, not processing map files. The actual results just live right here inside this diary log.

Things I still need to finish and track down:

- I still have not found a real old map image from the Imperial Gazetteer books or old atlases to use as a visual slide in my writing. Tracking down the district history solved the main problem, but it does not give me an actual picture file.
- My second-year greenness script is running on my computer right now. I will bring the data blocks in as soon as it stops downloading.
- The FRA data list and my field checks around Karanja Ghadge are still completely open, exactly like I noted before.

Entry 14

My year-2024 satellite files from my script `download_seasonal_scenes_year2.py` finally finished downloading. They use the exact same WRS-2 map box as before (path 144, rows 045 and 046). I set it for the monsoon kharif peak between September and November 2024 and the winter rabi pre-harvest block between January and March 2025. Both maps landed with a perfect 0% cloud cover again. I ran them through my old zero-pixel correction and NDVI steps from Entry 3 and 7. Then I calculated the seasonal greenness drop for each taluka all over again. I set these numbers right side-by-side with my original year data from Entry 7 (which covered the kharif 2025 and rabi 2025-26 cycle).

This is the exact double-check I wanted since Entry 7, and it held up beautifully. Look at the numbers from Year 1: my Berar talukas averaged a 0.325 greenness point drop from monsoon to winter, while the Central Provinces side averaged 0.255 (that is a clean 0.070 gap). Now look at Year 2, which is a completely separate year tracking an earlier cycle: the Berar side averaged a 0.316 drop, and the Central Provinces side hit 0.241 (creating a 0.074 gap). Both the direction (Berar always drops more) and the size of the gap (0.070 versus 0.074) match up almost exactly across two separate years that do not share a single satellite photo. This is a real seasonal crop signature. It is definitely not a random glitch that just happened to show up in one strange year. This was the whole point of running this extra test before trusting my Entry 7 results for my final paper.

Four specific talukas (Chandur Railway, Goregaon, Ner, and Digras) came back with zero good pixels this time. This happened because of tiny cloud or edge-map gaps in this particular photo pair that were not there in year 1. I left them as blank NaN spots instead of guessing or faking a number. My final comparison ran on 64 out of the 68 talukas instead of the full group, which changes absolutely nothing since the two years look incredibly steady anyway.

Files I made in this step:

- `03_Data_Processed/Vegetation/ndvi_postmonsoon_y2_300m.tif` and `ndvi_preharvest_y2_300m.tif` — These are my two new year-2 fixed greenness maps downsampled to 300 meters to save disk space.
- `03_Data_Processed/Vegetation/taluka_seasonal_ndvi_signature_both_years.csv` — My master comparison table showing both years side-by-side for all 68 talukas.
- `07_Outputs/year2_ndvi_comparison.png` — A final image file showing my comparison plots for the monsoon-to-winter greenness drop across both years and divisions.

Things I still need to finish and track down:

- I still have not found a real, old period map image to use as a visual slide in my final report. My FRA notes and the Karanja Ghadge field check with Sawardoh photos are also still completely open and sitting idle on my computer exactly like before.
- This gives me exactly two full years of data confirmation now. A third year would be super cool to add later on, but it is definitely not blocking my project anymore. This seasonal greenness signature is holding up incredibly well, so I am treating it as a proven, solid fact rather than a temporary guess.

Entry 15

Every remaining open thing from Entries 9 to 14 is either my Karanja Ghadge ground photos (still on hold) or not worth more time right now, so this entry is about building the actual dashboard inside `06_Streamlit`. I matched it to the rest of my portfolio instead of building from scratch, and every chart and table on it reads live from my actual CSV files — including the Phase 4 regression itself, which re-runs live on the page and matches my diary numbers to 3-4 decimal places. I left the Theoretical Foundations diagram and the Ground Truth page as placeholders for now since I have not finished the pieces they depend on. Files dumped this entry:

- ndvi_postmonsoon_y2_300m.tif and ndvi_preharvest_y2_300m.tif (downsampled the year-2 greenness maps to 300m to save space).
- taluka_seasonal_ndvi_signature_both_years.csv — master comparison sheets for all 68 talukas side-by-side.
- year2_ndvi_comparison.png — final box plots for the monsoon-to-winter drop comparisons.

Caught a nasty bug right before shipping this: pandas.Styler.applymap doesn't even exist in the newer pandas version installed here. It completely broke the p-value color highlighting on the regression page during my first live test. Quick hotfix: wrote a version-safe .map()/.applymap() fallback block so the script doesn't crash depending on the environment. Didn't just read the code back to guess if it worked—actually automated a full sweep of all 11 pages plus home using Streamlit's AppTest harness against the real CSV data to verify the outputs.

One honest gap: the taluka boundary file and my Phase 4 table don't share identical spelling for 3 of the 68 talukas, so those three don't show up on the interactive map yet. Wrote that down directly on the page instead of hiding it.

Things I still need to finish and track down:

- Diagram for Theoretical Foundations not generated yet.
- Karanja Ghadge/Sawardoh ground truth still on hold.

- Development Log PDF and the other three document exports not generated yet.

Entry 16

Two quick fixes after actually seeing Entry 15's dashboard live. I checked my page layout against all 7 sibling projects this time instead of just one, and cut it down from 11 pages to 9 to match the settled pattern — folding Climate Controls and my Findings & Conclusion text into the Spatial Discontinuity Regression page instead of keeping them separate, since climate was always a control variable, not a headline finding.

I also fixed the Central Provinces color: Mughal Green barely showed up on the dark navy background, so I swapped it for a brighter icy green that actually reads clearly, and fixed two spots where button text had gone unreadable against it.

Things I still need to finish and track down:

- Diagram for Theoretical Foundations not generated yet.
- Karanja Ghadge/Sawardoh ground truth still on hold.
- The four final document PDFs still not generated.

Entry 17

Finished the Theoretical Foundations diagram — four panels showing the land split, the NDVI leaf-physics sketch, the seasonal winter greenness drop chart, and a simple chain tracking how land laws lead to crops and then to a satellite signature — matching the visual style of the rest of my portfolio. Confirmed the cleanup script actually ran and the old stale page files are gone.

Built out the actual GitHub repo (THE-COTTON-LINE, used a hyphen instead of underscore) instead of just leaving a blank 'git init' stub. Set up a real .gitignore to block the 02_Data_Raw folder (no point uploading 23.2 GB of raw Landsat scenes and grids) along with the heavy intermediate .tif files. Requirements.txt was sitting completely empty since the first data pull, so populated that and threw together a basic README for the local setup instructions. Ran the remote push from my terminal. Hit a weird Windows permissions error where git refused to work on the D: drive due to some "dubious ownership" glitch—fixed it by whitelisting the safe.directory setting.

Things I still need to finish: My field ground checks around Karanja Ghadge are still completely on hold. Also, my four final document PDFs are now down to three based on my Entry 18 decisions.

Entry 18

My project roadmap for Phase 5 always said I must properly explain the institutional-economics books. I needed to focus on the big fight over how people labeled the Central Provinces instead of just dropping in a lazy name-check. So I sat down and read the famous Banerjee and Iyer (2005) paper directly. Their main point was that Indian districts that started under landlord-based tax systems have much worse farming results today than non-landlord districts. This stays true even a hundred years after the whole setup was canceled. Then I read the direct challenge by Iversen, Palmer-Jones, and Sen (2013). They showed that Banerjee and Iyer coded nearly the entire Central Provinces as a pure landlord setup. They basically lumped Malguzari right in with the heavy Bengal and Bihar landlord styles. But the real history records show that the 1863 Malguzari rules came

with real land maps and actual legal protection for the tenants. It was a completely different, much more mixed system.

When these new researchers recoded the Central Provinces districts using the old record files, six of Banerjee and Iyer's main mathematical findings completely lost their strength. On top of that, throwing the Central Provinces out of the data completely made the whole paper result disappear. This is directly important for my own project, not just a random related fact. These are the exact same six districts sitting on the Central Provinces side of my own map border line! This fight is the exact reason why I treated Malguzari as its own separate group from day one instead of mixing it into a generic landlord tag like the older paper did. I also brought in the Dell (2010) paper on Peru's mining system and the Keele and Titiunik (2015) paper on geographic maps. These are the best math examples for testing a specific historical border line using a distance-to-boundary variable. Both are real, verified, and perfectly on point for my spatial jump design instead of just a loose regional comparison.

I verified every single citation before using it instead of just trusting my memory. I went online and checked the exact AER volume and page numbers for the Banerjee-Iyer paper. Then I read the full Manchester working-paper text for the IPS recoding study to confirm their actual argument rather than just skimming the abstract. I also checked Beckert's *Empire of Cotton* book to verify that 8.5% Berar famine death number that I have been carrying in my project README file since day one. Finally, I cross-checked the original papers for WorldClim, SRTM, and Landsat to make sure those citations are 100% accurate.

Wrote up `CL_Research_Paper.md`, matching the standard layout the rest of my portfolio uses (Abstract, Introduction, Literature Review, Data & Methodology, Results, Discussion, Limitations, Future Work, Conclusion, References). Every number in it came straight from Entries 2 to 16 without recalculating anything, including the messy, uneven results like the farm worker numbers flipping from a trend difference at the corridor scale to a level jump at the full footprint — reported as an honest open mismatch, not smoothed over.

I reported my ground truth check as an open task, not a complete one. The field check around Karanja Ghadge and Sawardoh is still completely on hold. I wrote this down directly inside the Limitations and Future Work parts of my paper. I used my single taluka data point exactly for what it actually is (June 2026: ground heat 55.97°C, greenness 0.264, and 45.8% clear pixel coverage). It sits there as a lonely reference point, definitely not as a finished check.

Things I still need to finish and track:

- I still have not written the Project Report or the Executive Summary documents. Following my standing rules, those final files will only get written once the whole project (including my field ground checks) is completely finished.
- My field check around Karanja Ghadge and Sawardoh village is still completely on hold. It remains my job to start that part whenever I get the green light.

Entry 19

Read back the full Research Paper draft after Entry 18 and honestly, it felt incredibly stiff. Too much lazy "this study explores" phrasing dominating every sentence, and way too many repetitive "not X, but Y" structures stacked on top of each other. The abstract was also just one monster block of text doing too much at once. None of the actual math or citations were broken, but the voice was just dead—didn't sound like a human

wrote it. Did an end-to-end rewrite pass to force everything into active first-person ("I built," "I checked") where I'm making a manual judgment call. Split the abstract into three shorter, readable pieces, and cut the clunky contrast phrasing down to where it actually adds value. No numbers or data changed—this was purely a voice pass to kill the robotic edge.

Compiled the Development Log and the Research Paper as PDFs for the first time so both download buttons on the dashboard actually work now.

The Project Report and Executive Summary are still not built. My field check around Karanja Ghadge and Sawardoh village has not happened yet. My standing rule stays active: those final two documents only get written once the whole project is completely finished (I am definitely not writing them sooner just because the Research Paper and Dev Log are done). I chose to state this directly on the site rather than building fake placeholder versions just to make the file list look full.

Entry 20

Fixed the portfolio card's map link — it was pointing straight at the live Streamlit app, which doesn't match how the rest of my portfolio works. Exported the six interactive-map views (border side, greenness decline, ground heat, worker ratio, farmed land, distance to boundary) as standalone static HTML pages instead, matching the dropdown pattern the other cards use. Also trimmed my project card's text summary down to match the length of the other cards on the page.

Things I still need to finish: Turn on GitHub Pages for the repo so the six map links actually resolve. My Project Report, Executive Summary, and field ground photos remain exactly where Entry 19 left them.

Entry 21

I picked my field ground checks back up. I did not actually plan a whole separate trip down to Karanja Ghadge like my old roadmap said. The great thing is I already had the exact right data sitting inside a totally different project on my laptop. I run a weekly tracking setup with GPS maps and actual phone photos on my family's own land inside this exact same taluka (it is part of a separate folder named Roots & Reflectance, which has nothing to do with this project but uses the exact same physical ground). It already has a working Sentinel-2 NDVI and NDRE math code running. Two of those farm plots sit literally inside Karanja Ghadge village, and five more are in Sawardoh village just a few kilometers away. This gives me real GPS points, dated ground photos, and actual field-level satellite timelines for seven different farm boundaries. That is way more than the 2 or 3 plots my original plan asked for, plus I did not have to leave a second random dataset lying around completely unused on my disk.

Built it out as a proper local data matrix instead of hardcoding raw text on the app pages. Made karanja_sawardoh_field_ground_truth.csv under the Ground_Truth folder to track all 7 fields in long-format (matches the rest of the pipeline schema). Dumped eight highly compressed crop/soil photos (pushed down to under 700 KB each so the app doesn't lag) into the streamlit assets directory.

The scale gap note: this isn't a pixel-identical check against the June 2026 taluka data (LST 55.97°C, NDVI 0.264, 45.8% coverage). That baseline was Landsat at 30m, taluka-averaged; my field numbers are Sentinel-2 at 10m on the exact farm shapes, mostly on a different date. No point blurring that difference. What the numbers actually show: the two Karanja fields came back at NDVI 0.329 and 0.287 on 14 July 2026—plausible

enough next to the rougher taluka numbers from June. Canopy in the ground photos looks closed and healthy, which pretty much tracks what those values predict anyway.

What the data actually shows is simple. The two fields in Karanja Ghadge came back with greenness numbers of 0.329 and 0.287 on July 14, 2026. This makes total sense next to the rougher, wider taluka numbers I found earlier. Also, my ground phone photos taken at those exact matching coordinates show thick, green, and completely healthy plant leaves. This matches what those satellite numbers predict instead of fighting them. The five fields in Sawardoh village moved in that exact same green direction over the same timeline, which gives me great supporting evidence for my story.

One genuinely cool side-finding. The Sentinel-2 satellite pass on June 21 for Karanja Ghadge came back with zero good pixels for both of my farm fields. It was a complete cloud block, definitely not a coding error in my script. This means a second, completely separate satellite sensor lost its view over the exact same corner of the map footprint where Landsat's taluka map was already stuck at a tiny 45.8% clear data coverage. Two completely different satellite systems gave me the exact same conclusion. This specific patch of Wardha is just genuinely hard to see cleanly from space because of weather blocks. I chose to write this discovery directly into my site text instead of hiding it like an annoying mistake.

I updated my research paper to match these new facts. I rewrote Section 6.6 completely. It no longer says "Not Yet Complete" because I replaced that note with my actual ground check results. Of course, I kept the scale warning inside the text instead of quietly throwing it away. Inside Section 7, I deleted the task line about finishing the Karanja Ghadge and Sawardoh field check. The simple reason is that the task is finally completely done. I also changed the very last lines of both my abstract and my final conclusion section. They used to say that this field check was the only loose thread in my whole project. Now, they wrap up the whole story in the exact same honest way. The results are fully verified with a clear warning on scale, and I did not try to oversell anything.

Things I still need to finish: I need to build the Project Report and the Executive Summary documents next. My old standing rule said these two papers must wait until my ground truth checks actually closed out. Since that just happened today, they are the next logical items on my track list. My main README file, the homepage navigation labels inside `app.py`, and my final portfolio overview card still say "still open" for this task, so I must edit them to match the new timeline. Also, turning on GitHub Pages for my six interactive map links from Entry 20 is still sitting there on my manual account to-do list.

Entry 22

I closed out every single thing Entry 21 left open. Then I actually caught two real accuracy problems right on the ground truth website page itself once I saw it running live.

My Project Report and Executive Summary are finally finished. Every single number inside both files — the 0.070 and 0.074 greenness gaps, the corridor 0.310 versus 0.299 near-zero result, the full footprint interaction p-values of 0.0348 and 0.0105, the farm laborer corridor number of 0.0374 with its p=0.0361 jackknife range, and the Karanja Ghadge baseline facts — came straight from my research paper and my diary logs from Entries 9 to 21. I did not recalculate or round any numbers differently. Built both as PDFs and updated the README and the dashboard's download links to point to the finished files.

Then I actually looked at the page instead of just the data. My two Karanja fields grow soyabean and tur. My data table and phone photos both say so. But when I saw it running on a website called THE COTTON

LINE, showing a big block of soyabean and pigeon pea looked totally off. It was factually correct but felt wrong for the project theme. Luckily, my Sawardoh data has pure cotton fields right there in the same file. So I chose to swap the photo lineup to lead with that instead. I pulled the cotton close-up and soil texture photos right to the front, and I dropped the generic overview and tur pictures completely.

I went a step further after that. Sawardoh has five fields total, but only two of them (Field 3 and Field 5) grow pure cotton with nothing else mixed in. The other three fields mix in tur, orange trees, or turmeric. If Sawardoh is standing in as my main cotton proof, it should show specifically those two pure plots, not the whole village in general. So I cut both the photo choices and the NDVI/NDRE data table down to just Fields 3 and 5. I found their own field-specific overview photos and left the other three Sawardoh fields out of the main webpage view entirely (they are still sitting inside my raw CSV table on disk, I just do not load them onto this specific page).

My final pass: I reordered the layout and made it interactive. I flipped the page order so Sawardoh's pure cotton fields lead the view and Karanja follows right behind. It makes way more sense to put the actual cotton proof up front on a project named for cotton, while keeping Karanja fully present since it was the taluka I named in my original plan anyway. Both zones now carry their own soil photo right side by side with their crop close-up, leaving me with nine ground photos total across the two parts.

I also added a real interactive chart at the bottom using the Plotly library instead of a boring static image. This matches the exact design pattern I already used on my Cotton Signature NDVI page (I used `go.Figure`, my custom `plotly_layout` for the dark navy style, and `st.tabs` so people can flip between NDVI and NDRE lines easily). The chart plots all four featured fields (Sawardoh 3 and 5, Karanja 1 and 2) on the exact same axis with one line per farm plot. It is fully hoverable and zoomable. Karanja's missing 21 June date shows up honestly as a line that only starts later on 14 July. I chose to show this gap as a blank space rather than padding or faking the numbers with fake data.

I checked every single number against the CSV directly before shipping this step. For Field 3, the NDVI numbers read 0.2808 which I rounded to 0.281, and the NDRE read 0.1553 which became 0.155. For Field 5, the NDVI was 0.3277 which became 0.328, and the NDRE was 0.1957 which I made 0.196. Looking at Karanja Field 1, the NDVI showed 0.3288 which I rounded to 0.329, and the NDRE showed 0.1813 which became 0.181. Finally, for Field 2, the NDVI was 0.2874 which became 0.287, and the NDRE was 0.1599 which I made 0.160. Every single metric card on my website page matches my source CSV table perfectly down to three decimal places. There is absolutely no rounding drift or messy guessing anywhere in my work.

I also ran a complete syntax check and a full sweep to find any stale or placeholder text across every single page and document in this project. There is absolutely nothing left over from my old draft files. Nothing points at data that does not exist on my computer disk yet.

Files updated this pass:

- taluka_population_agri_stats.csv and taluka_landuse_stats.csv — expanded to cover 8 out of 11 districts with the extra eastern blocks added.
- phase4_fullfootprint_dataset.csv — updated full regression master table to hold all 4 outcomes across 67–68 talukas depending on the model.
- phase4_fullfootprint_all4_plots.png — final scatter plots showing full-footprint outcome trends with distance from the line.

Things I still need to finish: Turning on GitHub Pages for my six interactive map links from Entry 20 is the only thing left. It is just a one-time manual account setting inside the code repository, definitely not a coding change. Everything else this project set out to do—including my map boundary construction, both math regression scales, the field ground checks, the full document set, and now my fully updated Ground Truth page—is completely finished.